

Local currency systemic risk *

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Abstract

Emerging countries governments increasingly issue local currency denominated bonds and foreign investors have been increasing their holdings of these assets. By issuing debt denominated in local currency, emerging countries governments eliminate exchange rate risk. The growing stock of local currency government debt in the financial portfolios of foreign investors increase their diversification and exposure to fast growing economies. In this paper, we highlight some of the risks associated to this recent trend. First, we adopt the *CoVaR* risk-measure to estimate the vulnerability of individual countries to systematic risk in the market for local currency government debt. Second, we show that our country level estimates of vulnerability are higher the larger the share of local currency debt held by foreign investors. A version of the old adage "When New York sneezes, London catches a cold," that described the relationship between the stock markets in these two cities, still applies between individual emerging countries and the aggregate market for local currency government debt.

Keywords: CoVaR, emerging markets, local currency debt, contagion, systemic risk

JEL Classification: C58, F21, F34, G11, G15, G32

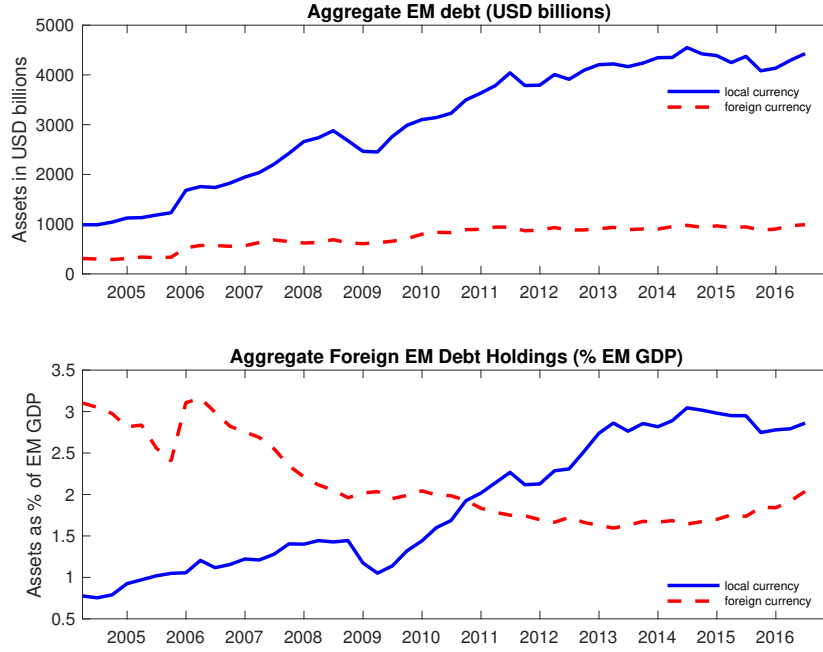
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1 Introduction

Emerging countries increasingly issue government bonds denominated in local currency and foreign investors share of this market has been progressively growing. By issuing debt in local currency, emerging countries governments can eliminate exchange rate risk which, instead, is typical of debt denominated in foreign currency (for example, [Eichengreen et al. \(2003\)](#) and [Margolin \(2007\)](#)). By investing in local currency denominated debt, foreign investors can increase the diversification of their financial portfolios and gain exposure to fast growing economies. In this paper, we highlight some of the risks associated to this recent trend. In fact, we show that the vulnerability of individual emerging countries with respect to systemic risk in the aggregate market for local currency debt increases with the share of local currency government debt in the hands of foreign investors. In some sense, a version of the old adage "When New York sneezes, London catches a cold", that used to describe the stock markets in these two cities, still applies between individual emerging countries and the overall market for local currency debt.

The stock of debt, issued by emerging countries governments and denominated in local currency, expressed in US dollars, increased by a factor of 4 since 2002 and it is currently worth over 4 trillions of US dollars. Over the same period, the stock of debt denominated in foreign currency was never above the 1 US dollar trillions mark (see figure 1). The growing stock of debt denominated in local currency shows that emerging countries are breaking free from the so called "original sin" (i.e, the difficulty emerging markets face when attempting to borrow abroad in their own currencies). Foreign investors are also progressively becoming more relevant players in the market for local currency debt. In the bottom panel of the figure 1, we report the foreign holdings of aggregate debt issued by emerging countries governments, as a percentage of emerging countries GDP. While the foreign holdings of debt in local currency have been progressively increasing, reaching approximately 3 percent of emerging countries GDP in 2016, the holdings of debt denominated in foreign currency have been declining and were approximately 2 percent in 2016. The increase share of local currency debt held by foreigners is a potential source of transmission of shocks both globally, and across different emerging markets. For example, [Kaminsky et al. \(2004\)](#) show that active trading strategies of equity mutual fund investors, like momentum strategies, spread contagion across emerging markets. [Beirne et al. \(2013\)](#) find volatility spillovers from mature to emerging stock markets. [Borri and Verdelhan \(2011\)](#) find that global factors explain a large share of the variability of returns on foreign currency denominated emerging countries government bonds. To the best of our knowledge, this is the first paper that studies the risk of contagion in the market for local currency government debt of emerging countries.

Figure 1: Aggregate Emerging Countries Government Debt



Notes: The top panel of this figure reports aggregate emerging countries government debt in US billions. The blue line is debt denominated in local currency. The dashed red line debt denominated in foreign currency. Local currency debt is converted in US dollars using the market exchange rate. The bottom panel of this figure reports the aggregate foreign investors holdings of emerging countries government debt denominated in local currency (blue line) and foreign currency (dashed red line). The aggregate holdings are reported as share of aggregate emerging countries government GDP. Data are from [Arslanalp and Tsuda \(2014\)](#) for the period 2002:Q1 – 2016:Q2 at quarterly frequency.

We estimate the vulnerability to systemic risk in the market for local currency debt with the reduced-form risk-measure *CoVaR*, first proposed by [Adrian and Brunnermeier \(2016\)](#)¹. Systemic risk measures the increase in tail comovement that can arise due to the spreading of distress across different financial markets. *CoVaR* is a measure of risk conditional upon an adverse shock, where risk is the standard value-at-risk (VaR). The VaR measures risk in terms of returns at a given probability: for example, a VaR of -10% at the 1% confidence level indicates that there is a probability of 1% of a return that is lower or equal to 10 percent. [Adrian and Brunnermeier](#) use *CoVaR* to estimate the systemic risk of financial institutions using a combination of market and balance sheet data. [Fong and Wong \(2012\)](#) adopt a similar approach to estimate bi-lateral systemic risk in the Eurozone using sovereign CDS data. In this paper, we find that emerging countries differ substantially in terms of their vulnerability to local currency debt systemic risk. In addition, countries with a larger

¹There exist several alternative measures of systemic risk. Many of them rely on CDS data. For example, [Acharya et al. \(2012\)](#) focus on high-frequency marginal expected shortfall; [Acharya et al. \(2017\)](#) and [Brownlees and Engle \(2016\)](#) develop SRISK, which measures capital shortfall conditional on market stress; [Billio et al. \(2012\)](#) builds a risk-measure based on Granger causality across institutions.

share of local currency debt held by foreign investors are more vulnerable suggesting that increasing integration of capital markets could bring contagion across emerging countries. These results have important implications for governments and financial institutions both in terms of optimal currency denomination of debt and risk management, and in terms of provisions to curb the volatility of foreign investors portfolio allocations. In addition, these results are valuable for foreign investors to correctly measure the risk and diversification of their financial portfolios as also argued by [Das and Uppal \(2004\)](#) for international equity portfolios.

This paper contributes to two strands of the literature. The first studies contagion in emerging markets. Contagion is usually defined as correlation between markets in excess of what would be implied by economic fundamentals ([Bekaert and Harvey, 2003a](#)). Most of the research in this strand of the literature has focused on equity markets, and on large financial and economic crises, like the 1994 Mexican and the 1997 Asian crises. [Calvo and Reinhart \(1996\)](#) find evidence of increased comovement across world equity and Brady bonds during the Mexican crisis and argue that contagion is more regional than global. [Calvo and Mendoza \(2000\)](#) is a classic model in which globalization may promote contagion by weakening incentives for gathering costly information. [Edwards and Susmel \(2001\)](#) find strong evidence of stock market volatility co-movements across countries. [Forbes and Rigobon \(2002\)](#) argue that it is important to correctly account for heteroskedasticity when measuring contagion with correlation coefficients and find, one hand, little evidence of contagion during major financial crises, and, on the other, a high level of market comovements in normal times. [Bekaert and Harvey \(2003b\)](#) estimate contagion as correlation across residuals of a two-factor model with time-varying betas applied to stock returns in three different regions – Europe, South-East Asia, and Latin America – and find no evidence of contagion from the Mexican crisis, but find evidence of contagion during the Asian crisis. [Tai \(2007\)](#) finds that Asian emerging stock markets become integrated into world capital markets since their official liberalization, and that there is evidence of contagion from stock to foreign exchange markets during the Asian crisis. [Dell’Ariccia et al. \(2016\)](#) study systemic risk for firms in emerging markets that borrow in foreign currency and that have, typically, income in local currency and argue for regulatory measures to limit currency mismatches. The second strand of the literature, studies the growing market for local currency sovereign and corporate debt in emerging countries. [Peiris \(2010\)](#) finds that higher foreign participation reduces long-term local currency government bond yields and volatility in a group of 10 emerging countries for the period 2000–2009. [Miyajima et al. \(2015\)](#) argue that local factors, rather than global factors, are the main drivers of local currency bond yields. [Du and Schreger \(2016\)](#) construct a measure of local currency credit spread as the yield difference between the rate on local currency

bonds and a synthetic local currency risk-free rate computed using cross-currency swaps. The authors find that the local currency spread is positive, and sizable and, contrary to foreign currency credit spreads, depend more on domestic, than global, conditions and have lower means and cross-country correlations. [Hale et al. \(2016\)](#) look at individual private, rather than sovereign, bonds issued in local currency. [Borri and Shakhnov \(2017\)](#) build a model with two partially segmented markets for local and foreign currency denominated government debt and show that the larger the share of foreign investors the higher the degree of integration across the two markets and globally.

The remainder of the paper is organized as follows. Section [2](#) describes the data. Section [3](#) introduces our conditional measure of vulnerability to systematic risk. Section [4](#) presents the results of our estimations. Section [5](#) concludes.

2 Data

We collect daily frequency total return indices of local currency debt issued by emerging governments from J.P. Morgan through Datastream. In particular, we use the J.P. Morgan GBI-EM indices for local currency denominated debt. The JPM GBI-EM indices track local currency bonds issued by emerging market governments. The countries in our sample are Argentina, Brazil, Chile, China, Colombia, Hungary, India, Indonesia, Malaysia, Perú, Philippines, Poland, Romania, Russia, South Africa, and Turkey. The longest sample is 31/12/2002–30/6/2016, but the size of the cross-section progressively increases as more countries enter the indices. In particular, there are 7 countries at the beginning of the sample and 17 at the end. Bonds included in the J.P. Morgan GBI-EM indices had an average maturity of 5 years at the end of 2015. J.P. Morgan also publishes two additional families of emerging market bond indices. The classic EMBI indices, formed in the early 1990s after the issuance of the first Brady bonds, collect foreign currency denominated debt (mostly U.S. dollar denominated). The ELMI+ indices also collect local currency debt, but for the money market and therefore have a very short duration (on average, 0.15 years). Note that J.P. Morgan includes in the indices individual bonds that meet specified criteria in terms of liquidity and reliability of market prices. For the countries in the sample, we additionally collect exchange data with respect to the U.S. dollar from Reuters and Barclays through Datastream (exchange rates are in units of foreign currency per U.S. dollar). We also collect daily frequency total return indices on the S&P500 Composite, the CBOE VIX, the Bank of America Merrill Lynch US Corporate Bond Index, and the Bloomberg Commodity Index from Datastream. Finally, we collect quarterly data on the shares of local currency and total government debt held by foreigners, nominal GDP, and shares of local currency and

total government debt over GDP from [Arslanalp and Tsuda \(2014\)](#) for the period 2002:Q1–2016:Q2, and the annual values of the index by [Chinn and Ito \(2006\)](#) (KAOPEN) measuring a country’s degree of capital account openness².

Table 1: Descriptive Statistics: financial data

	Panel A: local currency returns					Panel B: exchange rates	
	mean (%)	std.dev (%)	skew	min (%)	max (%)	mean (%)	std.dev (%)
Argentina	0.027	1.183	-1.185	-14.063	10.213	0.042	0.943
Brazil	0.048	0.316	-8.546	-9.626	2.174	-0.003	0.924
Chile	0.011	0.124	-0.127	-0.799	0.987	-0.002	0.620
China	0.015	0.213	-1.865	-3.968	3.434	-0.006	0.108
Colombia	0.038	0.292	-0.798	-2.927	2.355	0.001	0.700
Hungary	0.030	0.407	-0.983	-5.218	3.737	0.007	0.984
India	0.026	0.304	-0.756	-4.804	3.719	0.010	0.426
Indonesia	0.048	0.692	-1.007	-11.121	8.912	0.011	0.486
Malaysia	0.015	0.142	-0.558	-1.426	1.322	0.001	0.394
Mexico	0.034	0.325	0.518	-2.563	4.061	0.016	0.717
Peru	0.027	0.423	1.973	-3.209	7.012	-0.002	0.295
Philippines	0.027	0.433	-0.814	-3.806	2.577	-0.004	0.331
Poland	0.023	0.195	-0.841	-2.458	1.348	0.001	0.913
Romania	0.027	0.145	-0.737	-0.746	0.565	0.005	0.786
Russia	0.030	0.539	-2.007	-10.369	8.320	0.020	0.851
South Africa	0.034	0.399	-1.331	-6.516	2.819	0.015	1.079
Turkey	0.051	0.372	-0.496	-3.196	2.591	0.016	0.825
Mean	0.030	0.372	-1.151	-5.107	3.891	0.008	0.825
	Panel C: state variables						
S&P 500	0.032	1.186	-0.318	-9.460	10.958		
CBOE VIX	-0.016	6.725	0.729	-35.059	49.601		
U.S. Corporate	0.016	0.648	-0.546	-5.674	4.100		
Commodities	-0.002	1.085	-0.230	-6.401	5.650		

Notes: This table reports descriptive statistics on the financial data described in section 2. Panel A reports mean, standard deviation, skewness, minimum and maximum for the log returns on local currency government bond indices. Panel B reports mean and standard deviations for the log changes in exchange rate with respect to the U.S. dollar (exchange rates are expressed in units of local currency per U.S. dollar). Panel C reports mean, standard deviation, skewness, minimum and maximum for the log returns on the state variables used in the conditional estimation of *CoVaR* described in section 4. The sample is 31/12/2002–30/6/2016. Data are from Datastream at daily frequency.

We report in tables 1 and 2 descriptive statistics on the data. In panel A and C of table 1 we report mean, standard deviation, skewness, minimum and maximum values for the log

²The Chinn-Ito KAOPEN index is a number in between zero and one and it is equal to 1 for the U.S. For additional details on the data you can refer to the separate online appendix to this paper and to [Arslanalp and Tsuda \(2014\)](#) and [Chinn and Ito \(2006\)](#).

returns on the local currency bond indices and the state variables used in the estimation of $\Delta CoVaR$ of section 4. The mean daily returns in local currency are all positive and range from 0.011% for Chile to 0.051% percent for Turkey. Also volatilities differ substantially across countries: for example, in the period 31/12/2002–30/6/2016, Argentina registered a minimum daily return of -14% and a maximum daily return of 10%. Brazil, Indonesia and Russia also experienced large swings in the prices of their local currency government debt. Foreign investors buying bonds denominated in local currency also face exchange rate risk. In panel B we report the mean and standard deviation of the daily log changes in the exchange rates with respect to the U.S. dollar. For most countries, the mean exchange rate growth is positive indicating a depreciation of the local currency with respect to the dollar. Therefore, from the perspective of foreign investors, the investment in local currency bonds suffered a capital gain loss once converted back in dollars. In table 2 we report descriptive statistics on the share of local currency, and total, government debt held by foreigners, and on the local currency and total government debt over GDP. On average, foreign investors held 15 percent of emerging countries government debt denominated in local currency, and 30 percent of the total (i.e., local and foreign currency denominated). However, there exist both large differences across countries, and over time within a country. Foreign investors held only a small fraction of the local currency denominated debt issued by China and India (0.7% and 1.4% on average, respectively). On the contrary, foreign investors held a large fraction of local currency debt issued by countries like Hungary (average 30.2%, maximum 43.6%), Perú (average 32.0%, maximum 56.6%), Argentina (average 10.6%, maximum 40.3%), or Indonesia (average 22.8%, maximum 39.6%). Most of the government debt of emerging countries governments is denominated in local currency. On average, government debt in local currency is about 22 percent of GDP, while the total debt (i.e., local and foreign currency denominated) is only a bit larger and approximately equal to 30 percent.

3 Model

In this paper we follow [Adrian and Brunnermeier \(2016\)](#) and estimate $\Delta CoVaR$ with quantile regressions ([Koenker and Bassett Jr, 1978](#); [Koenker, 2005](#)). This is not the only possible estimation technique. For example, $\Delta CoVaR$ can also be estimated with generalized autoregressive heteroskedasticity (GARCH) models. We leave the interested read to the detailed discussion in [Adrian and Brunnermeier \(2016\)](#) for further details and proofs. We denote the log returns on the aggregate market for local currency debt issued by emerging countries with r^m . Define with Var_q^m the maximum market return at a confidence level of $1 - q$

$$Pr(r^m \leq VaR_q^m) = q. \quad (1)$$

Intuitively, VaR^m (i.e., the value-at-risk) corresponds to the maximum return in a bad state of the world, i.e., in a situation of aggregate distress for emerging countries. We define $CoVaR_q^{i|m}$ as the VaR of country i conditional upon the aggregate market m being in a state of distress (i.e., being at VaR^m)

$$Pr(r^i \leq CoVaR_q^{i|m} | r^m = VaR_q^m) = q \quad (2)$$

In order to estimate the conditional risk we use the following quantile regression

$$r_{t+1}^i = \beta_{0,q}^{i|m} + \beta_{1,q}^{i|m} r_{t+1}^m + \sum_{k=1}^K \gamma_{k,q}^{i|m} r_t^k + \epsilon_{t+1}^{i|m}, \quad (3)$$

where r^k are the lagged returns on a set of common factors that we use as conditioning variables. The coefficient $\beta_{1,q}^{i|m}$ measures how vulnerable country i is with respect to a state of distress of the aggregate market for local currency government debt issued by emerging countries. CoVaR is then obtained as fitted value of the quantile regression (3)

$$CoVaR_q^{i|r^m=VaR_q^m} = \beta_{0,q}^{i|m} + \beta_{1,q}^{i|m} VaR_q^m + \sum_{k=1}^K \gamma_{k,q}^{i|m} r^k, \quad (4)$$

where the values of the common factors r^k in (4) are those on the date when VaR_q^m is observed. We denote the marginal contribution of aggregate risk in the market for local currency government debt issued by emerging countries to the risk of country i with

$$\Delta CoVaR_q^{i|m} = CoVaR_q^{i|r^m=VaR_q^m} - CoVaR_q^{i|r^m=VaR_{0.5}^m}. \quad (5)$$

$\Delta CoVaR$ measures the difference between the $CoVaR$ of country i conditional on a state of aggregate distress in the market and the $CoVaR$ of country i conditional on the *median* state of the market (i.e., $q = 0.5\%$). Therefore, the larger the $\Delta CoVaR$, the higher the vulnerability of country i to systemic risk in the market for local currency emerging countries debt. Note that we use $\Delta CoVaR$ as a measure of vulnerability of individual countries. On the contrary, [Adrian and Brunnermeier \(2016\)](#) use $\Delta CoVaR$ to measure the systemic risk of individual financial institutions, i.e., the vulnerability of the entire financial market with respect to a state of distress of a single financial institution.

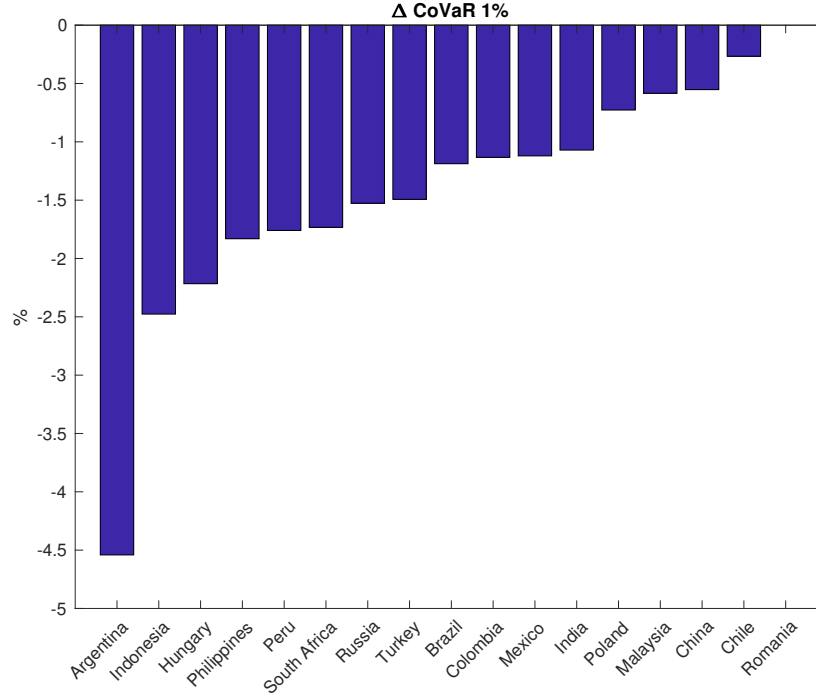
4 Estimation Results

We first estimate the conditional $\Delta CoVaR_{q,t}^i$ for country $i = 1, \dots$ and confidence level $1 - q$ over the full sample period 31/12/2002–31/12/2016 on daily frequency data³. As in [Adrian and Brunnermeier \(2016\)](#), we include in the estimation state variables that capture time variation in the conditional moments of asset returns. Including the state variables is important to disentangle the vulnerability of each country with respect to the aggregate market for local currency government debt from the that with respect to aggregate world factors. The set of state variables includes: the returns on the US market, proxied by the S&P500; the log changes in the CBOE VIX; the returns on a broad US corporate bond index and the returns on a index representing the commodity market. Note that the state variables are not aggregate risk factors, but rather variables that condition the mean and the volatility of the $\Delta CoVaR_{q,t}^i$. The returns on the aggregate market for local currency government debt is a simple average (i.e., equally weighted) of the individual countries returns. In the estimation of the $\Delta CoVaR_{q,t}^i$ for country i , we exclude country i from the set of countries used to build the aggregate market return r^m . We set the confidence level q to 1%. [Figure 2](#) plots the estimates of the conditional daily $\Delta CoVaR$. Argentina is by far the most vulnerable country, with a $\Delta CoVaR$ of about -4.5%. This means that, in a situation of particular distress in the market for local currency government debt, we should expect a maximum return on local currency Argentinian bonds, at the 1 percent confidence level, of -4.5%, after controlling for global state variables. Malaysia, China, Chile and Romania are the less vulnerable countries, all with daily $\Delta CoVaR$ larger than -1%. In [table 3](#) we report our estimates for the conditional and unconditional $\Delta CoVaR$ over the full sample. We also report bootstrap standard errors that show that our estimates are generally highly significant.

We next study the determinants of $\Delta CoVaR_{q,t}^i$, and in particular its possible relation with the share of local currency debt held by foreign investors. The intuition being that countries with a larger share of local currency government debt held by foreign investors are exposed to changes in their views and portfolio allocations (for example, [Aggarwal et al. \(1999\)](#); [Chari and Kehoe \(2003\)](#)). First, we repeat the estimation of the $\Delta CoVaR_{q,t}^i$ over non-overlapping quarters over the sample 2003:Q1–2016:Q3, using daily frequency data. Therefore, we obtain end-of-quarter day estimates for the conditional $\Delta CoVaR_{q,t}^i$, where we

³We focus on returns in local currency. [Brusa et al. \(2014\)](#) show evidence that foreign investors are compensated for bearing currency risk when investing in local stock markets. For most countries, the share of local currency government debt held by foreigners is small so that currency premia are less relevant. [Borri and Shakhnov \(2017\)](#) consider a model with partially segmented markets for local and foreign currency denominated government debt. We leave for future research the exploration between additional transmission channels in the market for local currency debt operating through changes in the exchange rate.

Figure 2: $\Delta CoVaR$ 1% (conditional)



Notes: In this figure we report the estimates for the weekly $\Delta CoVaR$ at the confidence level $q = 1\%$ for all the countries in the sample. Countries are sorted, from left to right, in order of smaller $\Delta CoVaR$. The sample period is 31/12/2002 – 31/12/2016, or the longest available for countries included in the J.P Morgan GBI-EM index after the beginning of our sample. $\Delta CoVaR$ is estimated using as state variables the returns on the S&P500, a U.S. corporate bond index, the CBOE VIX, and a commodity index.

use the values of the state variables on the dates when the VaR_q^m was observed. Second, we estimate a panel where the dependent variables are the $\Delta CoVaR_{q,t}^i$ (with $q = 1\%$) and we always include country fixed effect to account for county-level characteristics that are constant over the sample. Results are reported in table 4, where each column correspond to a different specification of the estimation equation. We start with a simple model where the only explanatory variables are the foreign holdings of local currency government debt. The estimated coefficient is negative and statically significant at the 1 percent level. Note that the estimated coefficient is also significant from an economic point of view: in fact, a 1 percentage point increase in the share of local currency debt held by foreigners is associated with a decrease in the daily $\Delta CoVaR_{q,t}^i$ by 2 percentage points, or with an increase in the loss conditional on a situation of distress in the market for local currency government debt. In the second and third columns we add first standard errors clustered at the country level, to account for possible auto-correlation and heteroskedasticity of the residuals, and second quarter time fixed effect, to account for common factors across countries that change over

time. The inclusion of the time fixed effect is important as it improves the explanatory power of the regression and increases, in absolute value, the coefficient attached to the share of local currency debt held by foreigners to -5%. In columns 4 to 7 we experiment adding additional explanatory variables. We find that the size of the economy (log GDP), the ratios between local currency and total government debt to GDP, and the share of total government debt held by foreign investors are all not statistically significant from zero. We also include the Chinn-Ito KAOPEN index, measuring a country's degree of capital account openness. A higher index number corresponds to a more open and integrated economy. However, also in this case the estimated coefficient is not statistically significant. Finally, we include the quarterly growth rate in the exchange rates with respect to the U.S. dollar (an increase in the exchange rate implies a depreciation of the local currency). The coefficient is negative, statistically significant, and of a magnitude similar to that on the share of local currency debt held by foreign investors.

5 Conclusions

Emerging countries increasingly issue government bonds denominated in local currency and foreign investors share of this market has been progressively growing. By issuing debt in local currency, emerging countries governments can eliminate exchange rate risk which, instead, is typical of debt denominated in foreign currency. By investing in local currency denominated debt, foreign investors can increase the diversification of their financial portfolios and gain exposure to fast growing economies. However, foreign investors could act as a channel of transmission of shocks, globally and across emerging markets. In this paper we focus on the latter possibility, and find that the share of local currency debt held by foreign investors positively affects the vulnerability of individual countries to systemic risk in the market for local currency government debt. These results have interesting implications for governments and financial institutions both in terms of optimal currency denomination of debt and risk management, and in terms of provisions to curb the volatility of foreign investors portfolio allocations. In addition, these results are important for foreign investors in correctly measure the risk of their financial portfolios. We leave for future research an in-depth analysis of the policy implications of these results.

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Table 2: Descriptive Statistics: debt and macro data

	for. holdings local cur. debt			for. holdings total debt			local debt to GDP			total debt to GDP		
	mean	min	max	mean	min	max	mean	min	max	mean	min	max
%												
Argentina	10.6	0.0	40.3	38.9	21.4	57.6	9.9	1.8	16.0	32.0	23.7	54.3
Brazil	10.9	1.0	20.3	11.6	7.9	16.3	41.8	36.6	47.1	61.3	55.2	70.1
Chile	5.2	3.1	8.9	35.5	12.9	93.8	6.3	0.5	16.7	8.2	2.6	20.3
China	0.7	0.0	3.2	1.5	0.4	4.0	14.0	12.3	17.0	16.1	14.3	19.3
Colombia	11.5	4.5	22.3	29.7	23.3	51.0	24.1	18.3	27.8	30.6	28.1	34.8
Hungary	30.2	21.0	43.6	47.6	39.5	56.0	41.6	34.7	49.4	59.7	50.4	67.9
India	1.4	0.1	4.0	1.4	0.1	4.0	31.2	27.8	35.3	31.2	27.8	35.3
Indonesia	22.8	1.6	39.6	35.3	4.6	61.6	12.1	9.2	18.9	14.6	11.6	19.6
Malaysia	25.0	12.4	37.3	28.0	14.3	37.0	40.7	32.2	50.6	42.2	33.7	50.3
Mexico	20.4	4.4	38.0	38.1	22.9	60.0	22.0	11.9	32.6	22.3	16.4	28.9
Peru	32.0	0.0	56.6	52.2	39.9	68.7	5.6	1.3	8.4	15.2	12.4	20.0
Philippines	6.0	0.0	14.6	27.5	23.3	34.7	32.2	27.7	38.9	35.3	28.3	46.9
Poland	26.7	13.3	41.9	43.3	31.8	59.0	32.3	27.8	36.2	41.5	34.7	47.2
Romania	15.9	5.4	22.4	43.5	22.2	73.9	13.7	8.2	15.8	14.6	3.5	27.5
Russia	7.3	0.1	23.1	29.0	17.9	50.3	5.1	3.4	7.2	8.5	5.8	12.6
South Africa	23.7	6.5	38.0	25.4	14.4	38.2	26.7	18.3	38.1	34.6	23.8	48.6
Turkey	14.7	5.1	24.5	25.2	13.3	37.2	28.5	22.3	34.6	36.6	28.2	53.5
Mean	15.6	4.6	28.2	30.2	18.2	47.3	22.8	17.3	28.9	29.7	23.6	38.7

Notes: This table reports mean, min and max values for the share of local currency government debt held by foreigners, the share of total government debt (i.e., local and foreign currency denominated) held by foreigners, the ratio between local currency government debt and GDP, the ratio between total government debt and GDP. The sample is 31/12/2002-30/6/2016. Data are from [Arslanap and Tsuda \(2014\)](#) at quarterly frequency.

Table 3: Full sample $\Delta CoVaR$ 1%

	$\Delta CoVaR$ 1% conditional	s.e	$\Delta CoVaR$ 1% unconditional	s.e
Argentina	-4.541	2.387	-4.323	1.681
Brazil	-1.188	0.203	-1.142	0.127
Chile	-0.266	0.117	-0.338	0.054
China	-0.552	0.150	-0.590	0.078
Colombia	-1.133	0.239	-1.211	0.115
Hungary	-2.216	0.639	-1.726	0.194
India	-1.071	0.279	-0.957	0.068
Indonesia	-2.477	0.946	-2.853	0.529
Malaysia	-0.584	0.121	-0.563	0.061
Mexico	-1.120	0.266	-1.178	0.155
Peru	-1.760	0.608	-1.680	0.179
Philippines	-1.831	0.765	-2.028	0.366
Poland	-0.727	0.188	-0.706	0.088
Romania	-0.580	0.118	-0.700	0.057
Russia	-1.526	0.617	-1.707	0.234
South Africa	-1.733	0.422	-1.351	0.103
Turkey	-1.493	0.278	-1.648	0.131

Notes: In this table we report estimates for the $\Delta CoVaR$ with 1% confidence over the full sample 31/12/2002 – 30/6/2016 using daily frequency returns. The conditional $\Delta CoVaR$ is estimated using as lagged state variables the returns on the S&P500, a U.S. corporate bond index, the CBOE VIX, and a commodity index. Standard errors are by bootstrap. All numbers are reported in percentages.

Table 4: Determinants of $\Delta CoVaR$

VARIABLES	(1) $\Delta CoVaR$	(2) $\Delta CoVaR$	(3) $\Delta CoVaR$	(4) $\Delta CoVaR$	(5) $\Delta CoVaR$	(6) $\Delta CoVaR$	(7) $\Delta CoVaR$
foreign holdings local cur. debt	-0.02*** (0.01)	-0.02 (0.01)	-0.05* (0.03)	-0.05* (0.02)	-0.05* (0.03)	-0.04* (0.02)	-0.05* (0.02)
log GDP				0.01 (0.01)	0.01 (0.01)	0.01 (0.01)	0.00 (0.00)
foreign holdings total debt					0.01 (0.02)	-0.00 (0.02)	-0.00 (0.02)
local cur. debt to GDP				-0.00 (0.04)	-0.00 (0.04)	0.01 (0.04)	0.01 (0.04)
total debt to GDP					-0.02 (0.03)	-0.04 (0.03)	-0.04 (0.03)
Chinn-Ito Index						-0.01 (0.01)	-0.01 (0.01)
Δs							-0.03* (0.02)
Observations	683	683	683	683	683	683	683
R-squared	0.01	0.01	0.26	0.27	0.27	0.28	0.29
Number of countries	17	17	17	17	17	17	17
Country FE	YES	YES	YES	YES	YES	YES	YES
Year FE	NO	NO	YES	YES	YES	YES	YES
Clustered SE	NO	YES	YES	YES	YES	YES	YES

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Notes: In this table we report results of the panel estimation. The dependent variables are always the $\Delta CoVaR$ of country i , with confidence level 1%, conditional on the overall market for local currency debt issued by emerging markets being in a situation of distress (i.e., at VaR^m with confidence level 1%) and estimated using as state variables the returns on the S&P500, a U.S. corporate bond index, the CBOE VIX, and a commodity index. The $\Delta CoVaR$ are computed over non-overlapping quarters from 2003:Q1 to 2016:Q3. The values of the state variables are those on the dates when the VaR^m has been observed for every quarter. The independent variables are: foreign holdings of local currency government debt; log of GDP; local currency government debt as a share of GDP; total government debt as a share of GDP; the Chinn-Ito Index; the growth rate in the exchange rate with respect to the U.S. dollar. Standard errors are clustered at the country level. Data are from Datastream, [Arslanalp and Tsuda \(2014\)](#) and [Chinn and Ito \(2006\)](#).